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Project Title

**Machine learning based Buyer
Segmentation and Investment
Profiling for Real Estate Market
Intelligence**

Domain: Financial Analytics & Real Estate Market Intelligence

1. Why I did this project

Real estate companies collect a huge amount of information about the people who buy from them: their age, where they live, how they paid, how many properties they have bought, and more. Most of the time, this information just sits in a database and nobody looks at it closely.

I wanted to find out if this data could tell a useful story. Could I group buyers into a small number of types, based only on how they actually behave, and use those types to help the marketing and sales teams work smarter? That is the question this project answers.

To do this, I used a method called clustering. Clustering is a way of letting a computer find natural groups in data on its own, without anyone telling it in advance what the groups should look like. I only told the model how many groups to try to find, and it worked out which buyers belonged together.

2. The data I used

I worked with two tables of data.

- A client table with 2,000 buyers. Each row is one person, with details like age, gender, country, region, whether they applied for a loan, why they were buying, and how they found the property.
- A transaction table with 10,000 property records. Of these, 7,305 were already sold and 2,695 were still available (not yet sold).

Before doing anything else, I checked the quality of this data. The client table had zero duplicate rows and zero missing cells, which is a strong starting point. It means I did not need to spend time cleaning up broken or repeated records.

There was one detail worth explaining. Of the 10,000 property records, 2,695 available listings had no buyer linked to them yet, because nobody had bought them. Rather than guessing which buyer might eventually purchase these, I left them unassigned. Guessing would have made the results look more complete, but it would have been dishonest to the data.

3. How I built the model

Turning raw records into buyer-level features

A clustering model cannot read a table the way a person does. It needs numbers that describe each buyer in a consistent way. So the first step was feature engineering, which simply means turning raw data into useful numbers.

For every buyer, I calculated: their age, how many properties they had completed a purchase on, their total spending, their average price paid, the average size of the properties they bought, the average price per square foot, and what share of their purchases were office space rather than homes.

Preparing the data for the model

Some of the information I had, like loan status, buying purpose, and how they found the listing, was written in words, not numbers. I converted each of these into a set of yes-or-no columns, a method called one-hot

encoding. For example, instead of one column called "referral channel" with the word "Website" in it, the model sees a column called "Website" with a 1 if that is true and a 0 if it is not.

I also standardised all the numeric columns. This means putting every number on the same scale, usually centred around zero. Without this step, a column like total sales, which can be in the hundreds of thousands, would completely overpower a smaller column like age, simply because its numbers are bigger, not because it matters more.

Grouping the buyers

With the data ready, I used a method called K-Means clustering. In simple terms, K-Means picks a number of group centres, places each buyer with the nearest centre, then keeps adjusting the centres until the groups stop changing. The buyers inside one group end up looking similar to each other, and different from buyers in the other groups.

I also ran a second method, hierarchical clustering, purely as a check. It builds groups in a different way, by joining the closest buyers together step by step. I used it to confirm that the groups K-Means found were not a fluke of that one method, but a pattern that showed up again with a different approach.

4. How I picked the number of groups

K-Means needs to be told in advance how many groups to look for. That number is called K. Picking the right K matters: too few groups and you lose useful detail, too many and the groups stop meaning anything real.

I tested every value of K from 2 to 8 and compared them using two measures.

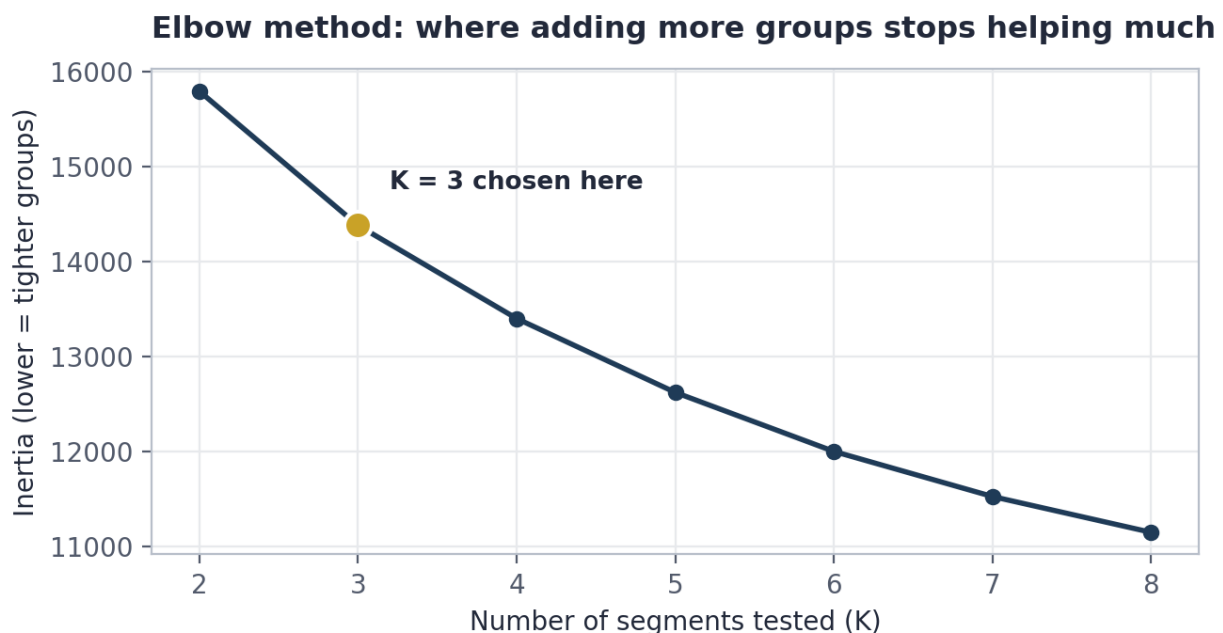


Figure 1: The elbow method. Inertia measures how tightly packed each group is. It always falls as K rises, so I looked for the point where it stops falling quickly.

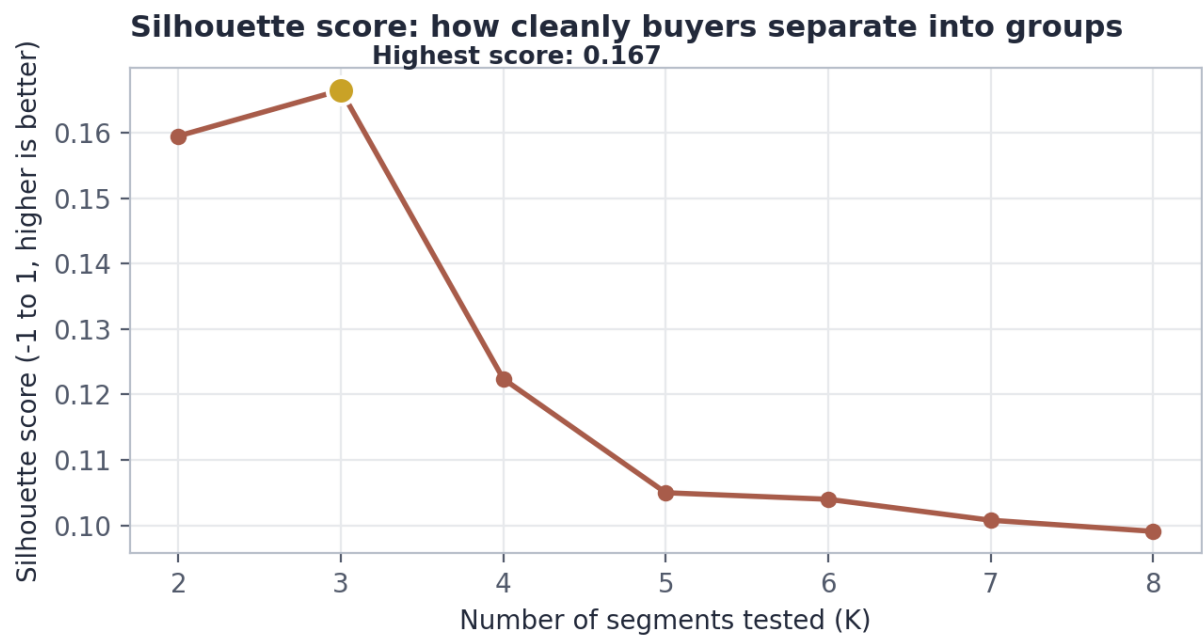


Figure 2: The silhouette score for each K. This score checks how well each buyer fits inside their own group compared to the next closest group. It runs from -1 to 1, and higher is better.

The silhouette score peaked at K=3, with a score of 0.167, the highest of every group size I tested. After K=4, the score drops sharply, which tells me the extra groups were splitting buyers apart without any real reason behind it.

K (segments)	Silhouette score	Inertia
2	0.1595	15,795
3 (chosen)	0.1665	14,383
4	0.1224	13,403
5	0.1050	12,625
6	0.1040	12,004
7	0.1008	11,529
8	0.0991	11,152

5. The three buyer segments I found

Three clear groups came out of the model. I named them by what actually sets them apart: how often a buyer purchases, and how much they typically spend.

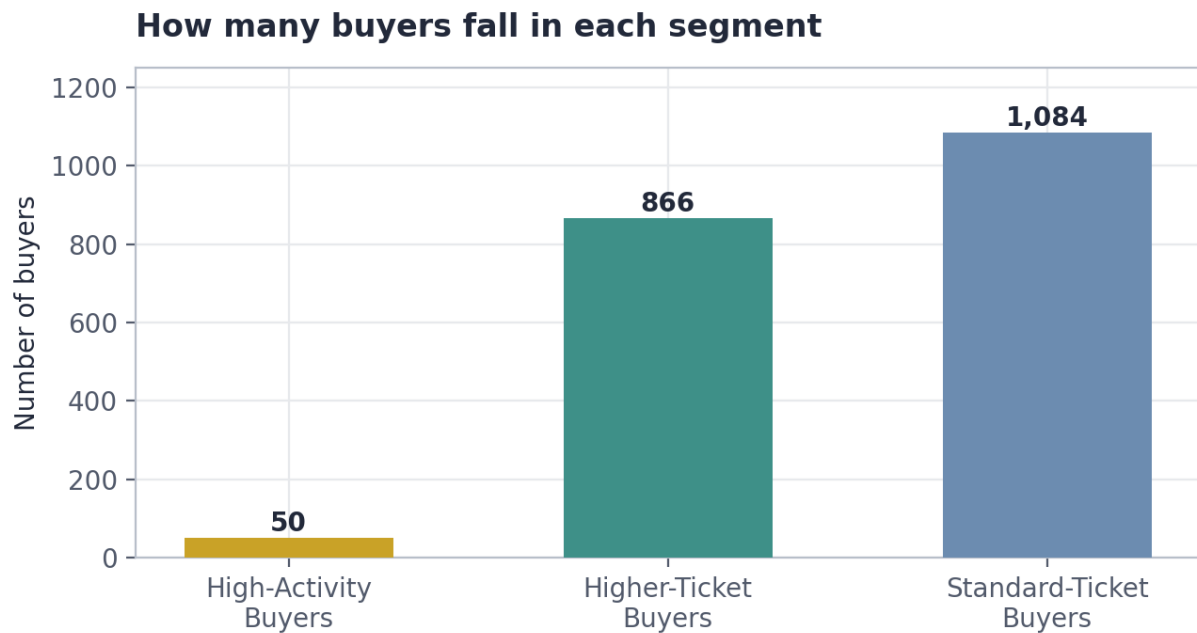


Figure 3: How the 2,000 buyers split across the three segments.

Segment	Buyers	Avg. purchases	Avg. sale price	Avg. total sales
High-Activity Buyers	50	7.32	\$332,747	\$2,416,603
Higher-Ticket Buyers	866	3.58	\$409,704	\$1,455,443
Standard-Ticket Buyers	1,084	3.54	\$297,730	\$1,051,206

High-Activity Buyers. Only 50 buyers, but they stand apart from everyone else. On average they complete 7.32 purchases each, more than double any other group, and their average total spending is over \$2.4 million. This is a small, valuable group worth knowing by name.

Higher-Ticket Buyers. The middle group in size, 866 buyers, but the top group in price. They pay the most per property on average, \$409,704, even though they buy less often than High-Activity buyers.

Standard-Ticket Buyers. The largest group, 1,084 buyers, and the most typical. Their purchase frequency and spending are both the lowest of the three groups, which makes them the baseline the other two segments stand out from.

6. How each segment behaves

Knowing how many properties a buyer buys and how much they spend is useful, but it does not say why they buy, how they pay, or how they found the listing. So I looked at these behaviours across the three segments.

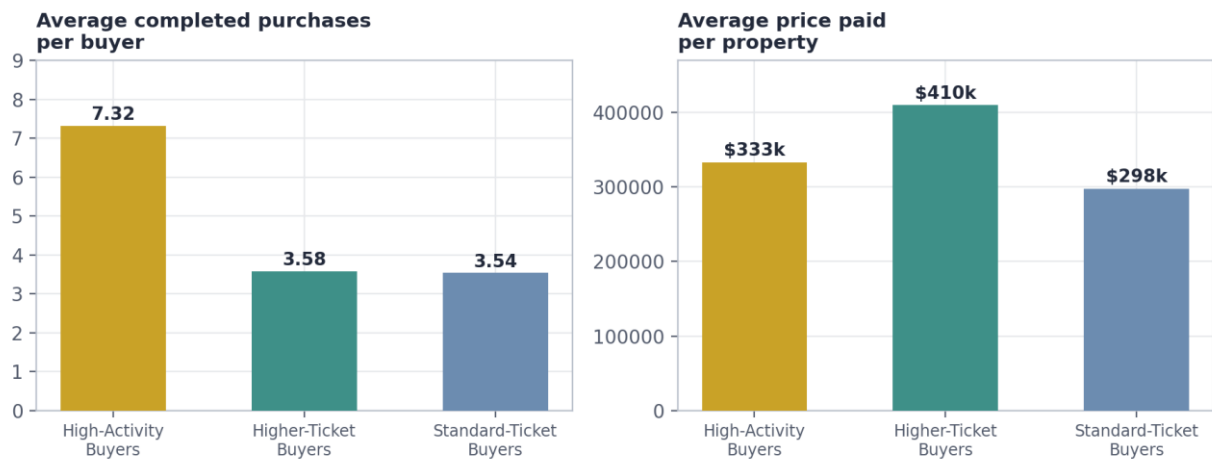


Figure 4: Average number of purchases and average price paid, side by side, for each segment.

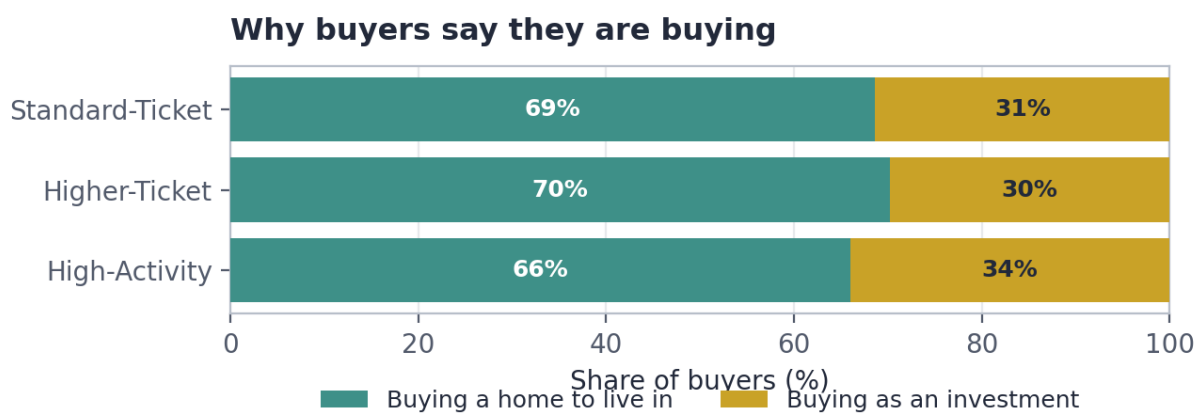


Figure 5: Whether buyers said they were purchasing a home to live in, or as an investment.

Across all three segments, most buyers are purchasing a home to live in rather than as an investment. High-Activity Buyers have the highest investment share at 34%, which fits their pattern of buying more often.

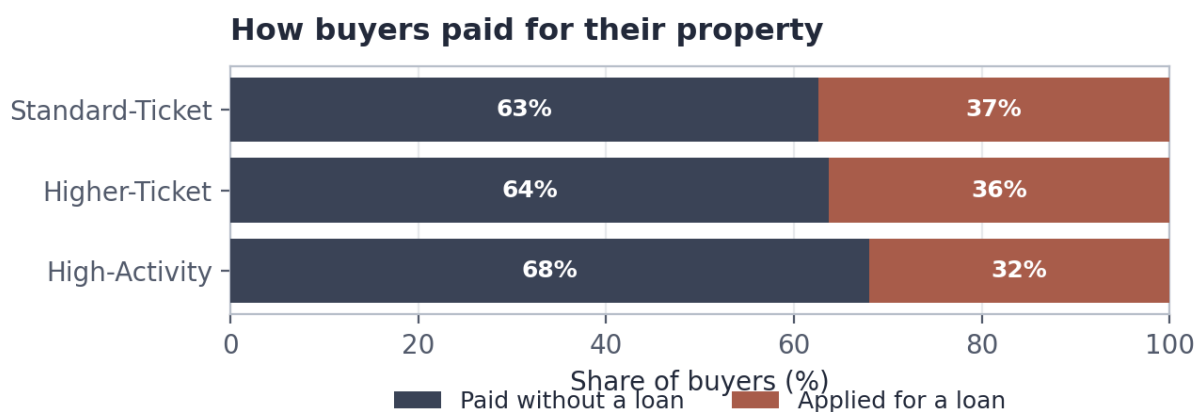


Figure 6: Whether buyers applied for a loan to finance their purchase.

Standard-Ticket and Higher-Ticket Buyers are more likely to use a loan (around 36 to 37%) than High-Activity Buyers (32%). This makes sense: buyers who purchase often are more likely to already have the funds available.

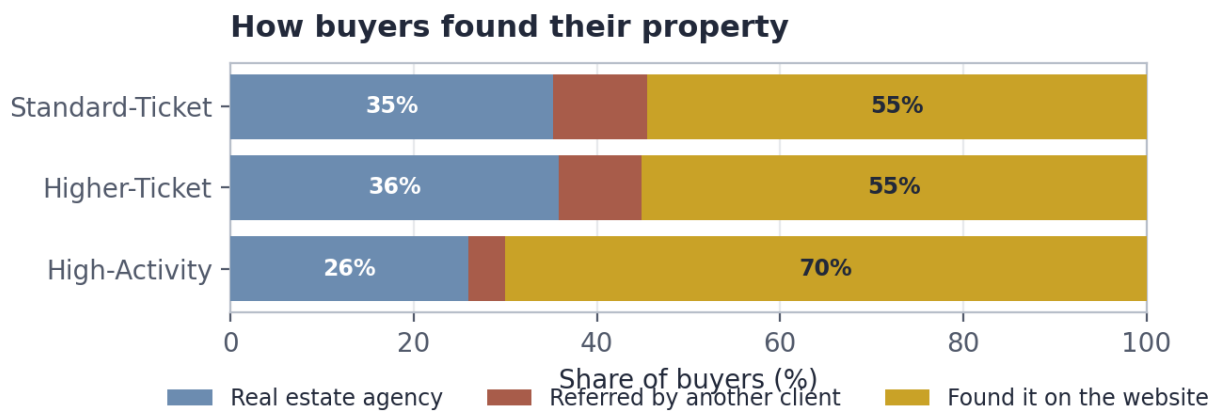


Figure 7: How buyers first found the property, split by channel.

The website is the single biggest channel for every segment, but it matters most for High-Activity Buyers, at 70%. Agency referrals and word-of-mouth from other clients play a bigger role for the other two segments.

7. Age and satisfaction across segments

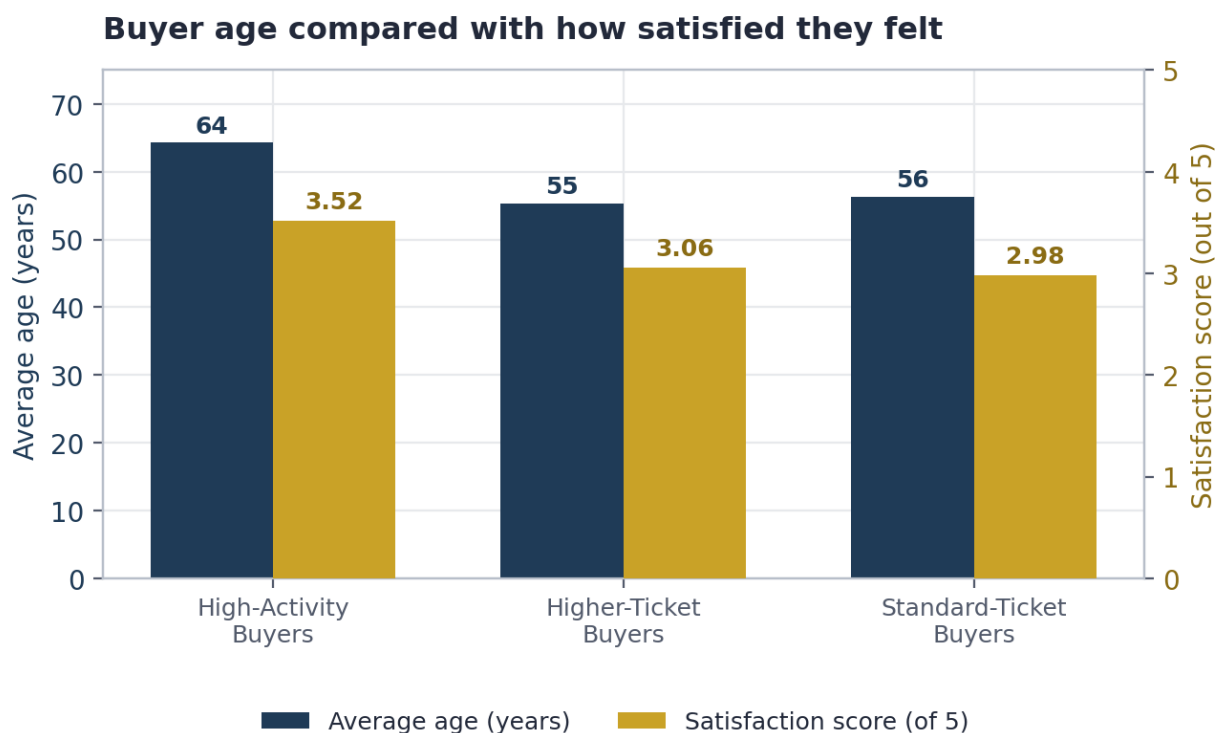


Figure 8: Average buyer age next to average satisfaction score, for each segment.

High-Activity Buyers are noticeably older on average, 64 years, compared to 55 and 56 years in the other two groups. They also report the highest satisfaction score, 3.52 out of 5. Standard-Ticket Buyers report the lowest satisfaction, 2.98 out of 5, though the gap between all three groups is small enough that I would not treat it as a major warning sign on its own.

8. Where these buyers live

The sample of buyers comes from ten countries, but it is heavily concentrated in one.

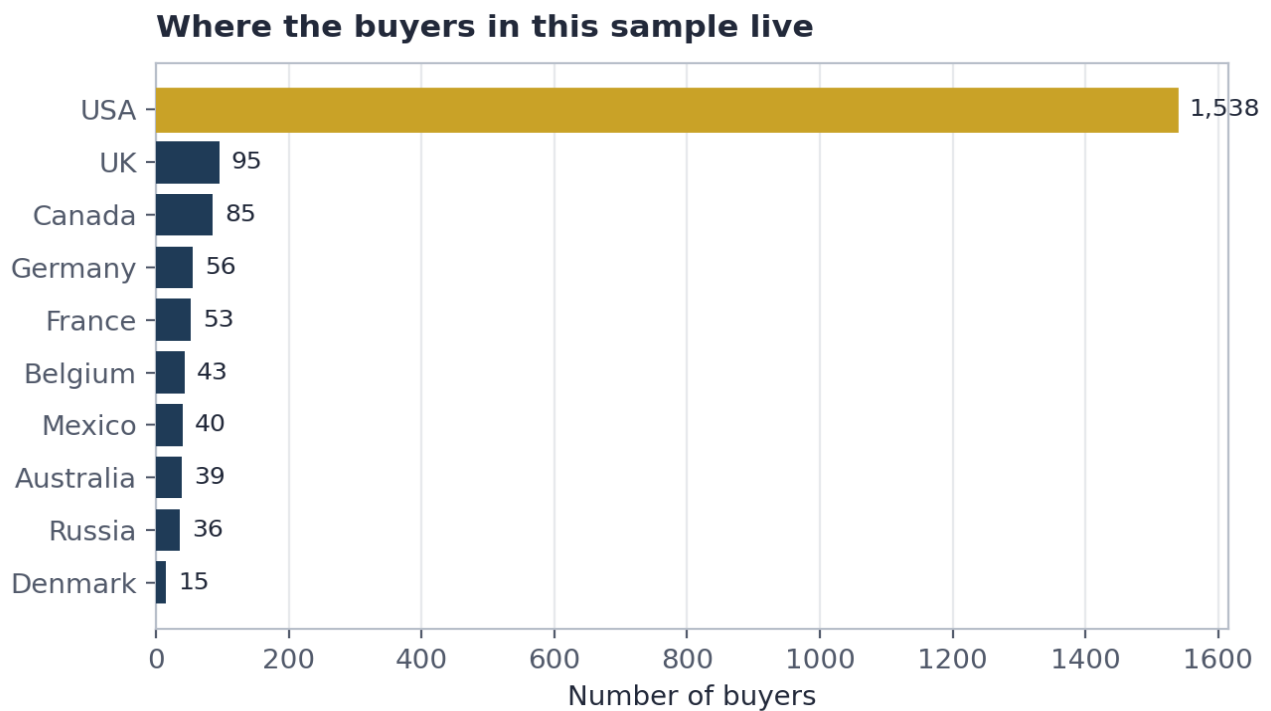


Figure 9: Number of buyers by country, across the full sample.

Country	Buyers	Share of sample
USA	1,538	76.9%
UK	95	4.8%
Canada	85	4.3%
Germany	56	2.8%
France	53	2.7%

The United States accounts for more than three out of every four buyers in this sample, and California is the single largest region within it. Because of this imbalance, I would be careful about drawing conclusions for other countries. There simply are not enough buyers from Germany, France, or Denmark yet to say anything reliable about them.

9. What I recommend

- Build separate marketing messages for each segment based on how often they buy and how much they spend, instead of sending everyone the same message.
- Track the website, agency referrals, and client referrals separately, and watch how conversion and repeat activity differ by segment and by channel.
- Create a financing-focused communication path for buyers who apply for loans, and a different path for buyers who do not.
- Use country and region as a targeting layer only after segmentation is applied, not before, and avoid assuming any market is more attractive without real data behind it.

- Collect more data in future: income, net worth, investment amount, how many properties a buyer already owns, and property location coordinates would all sharpen this kind of analysis considerably.
- Check every action taken on these segments against real business results: lead-to-sale conversion, repeat purchase rate, average ticket size, and campaign return on investment.

10. Limits of this study

I want to be upfront about what this analysis can and cannot tell us.

- These segments describe what buyers have done, not what they will do next. They are descriptive, not predictive.
- The data has no information on income, net worth, or investment amount, so I cannot label any segment by wealth. Any such label would be a guess dressed up as a fact.
- The silhouette score of 0.167 means the three groups are real but not perfectly separated. Some buyers sit close to the border between two segments. I would treat the segments as a useful starting point, not a rigid box.
- Available (unsold) listings without a matching buyer were left out of the buyer-level analysis, rather than guessed at.

11. Conclusion

This project turned 2,000 raw buyer records into three segments that a marketing or sales team could act on tomorrow: a small, high-value group of frequent buyers, a larger group that pays the most per property, and the largest group that represents the typical buyer. None of these groups were invented; they came directly out of how buyers actually behaved in the data.

The next useful step is not more modelling, it is testing. Try a segment-specific message on a small group of Higher-Ticket Buyers, measure what happens, and use that result to decide whether to expand it. That is how this analysis earns its value: not by sitting in a report, but by changing what the business does next.